

ENERGY & UTILITIES

AI-Powered Smart Grid Digital Twin

Intelligent Operational Ecosystem for Energy & Utilities.

LIVE SIGNALS → SIMULATED DECISIONS → SAFER ACTION

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THE OPERATING GAP

The grid is data-rich but decision-poor

Unexpected outages

Equipment failures interrupt supply.

Renewable uncertainty

Solar and wind change the operating picture.

Slow fault detection

Manual inspections delay response.

Fragmented grid data

Energy losses stay hard to locate.

Signals arrive, decisions lag.



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SIMULATE BEFORE YOU ACT

One digital twin predicts problems before they happen

01	02	03	04	05	06
Sensors	IoT/SCADA	AI Engine	Digital Twin	Prediction	Action
Capture	Connect	Analyze	Simulate	Anticipate	Respond



Every AI action is simulated in the twin first.

Failed simulations never reach the grid.

A safety gate between recommendation and execution.

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CONNECTED ARCHITECTURE

From live data to action in five layers

Data sources	SCADA · AMI smart meters · PMU · Weather · Solar/wind · GIS · Drone/thermal
Data platform	IoT platform · Time-series database · OT historian
AI engine	Forecasting · Anomaly detection · Predictive maintenance · Outage prediction · Optimization · Computer vision
Digital twin	Real-time virtual replica of the grid
Applications	Control room · Field engineer app · Executive analytics

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Six AI modules, one ranked decision, tested in the twin

SIX MODULES

Load and renewable generation forecasting

Fault and anomaly detection

Predictive maintenance and outage prediction

Power flow and DER optimization

Rank the intervention.
Test it in the twin.

SIMULATED SCENARIO

Transformer health anomaly

Live grid state · Fault location flagged

Without AI

Fault → Failure → Outage → Long downtime

With AI

Anomaly → Prediction → Engineer alert → Preventive maintenance

Scenario outcome: no major outage

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HUMAN-GUIDED AUTONOMY

Reactive grid → Predictive grid → Autonomous grid

Respond after failure → intervene before failure → execute validated actions.

Control room and field engineer app share one live grid state.

TARGETS, NOT MEASURED RESULTS

- Shorter outages, better prediction accuracy and faster field response
- Lower maintenance cost and energy losses
- Higher renewable utilization; more assets monitored at scale

AI · IoT · Digital Twin · Renewable Energy · Human Engineers

Predict. Prevent. Optimize. Power the Future.

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